LEMMA KILL COLIMIT WEAK GENERAL

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

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Lemma 0.1. *Let I, J be ideals of a Noetherian ring. Let M be a finite A-module. Let s and d be integers. With T as in (0EFG) assume*

1. *A has a dualizing complex,*
2. *if $\mathfrak{p} \in V(I)$, then no condition,*
3. *if $\mathfrak{p} \notin V(I)$, $\mathfrak{p} \in T$, then $\dim((A/\mathfrak{p})_\mathfrak{q}) \leq d$ for some $\mathfrak{q} \in V(\mathfrak{p}) \cap V(J) \cap V(I)$,*
4. *if $\mathfrak{p} \notin V(I)$, $\mathfrak{p} \notin T$, then*

$$\mathit{depth}_{A_\mathfrak{p}}(M_\mathfrak{p}) \geq s \quad \text{or} \quad \mathit{depth}_{A_\mathfrak{p}}(M_\mathfrak{p}) + \dim((A/\mathfrak{p})_\mathfrak{q}) > d + s$$

for all $\mathfrak{q} \in V(\mathfrak{p}) \cap V(J) \cap V(I)$.

Then there exists an ideal $J_0 \subset J$ with $V(J_0) \cap V(I) = V(J) \cap V(I)$ such that for any $J' \subset J_0$ with $V(J') \cap V(I) = V(J) \cap V(I)$ the map

$$R\Gamma_{J'}(M) \longrightarrow R\Gamma_{J_0}(M)$$

induces an isomorphism in cohomology in degrees $\leq s$ and moreover these modules are annihilated by a power of $J_0 I$.

Proof. Let us consider the set

$$B = \{\mathfrak{p} \notin V(I),\ \mathfrak{p} \in T,\ \text{and } \operatorname{depth}(M_\mathfrak{p}) \leq s\}$$

Choose $J_0 \subset J$ such that $V(J_0)$ is the closure of $B \cup V(J)$.

Claim I: $V(J_0) \cap V(I) = V(J) \cap V(I)$.

Proof of Claim I. The inclusion $\supset$ holds by construction. Let $\mathfrak{p}$ be a minimal prime of $V(J_0)$. If $\mathfrak{p} \in B \cup V(J)$, then either $\mathfrak{p} \in T$ or $\mathfrak{p} \in V(J)$ and in both cases $V(\mathfrak{p}) \cap V(I) \subset V(J) \cap V(I)$ as desired. If $\mathfrak{p} \notin B \cup V(J)$, then $V(\mathfrak{p}) \cap B$ is dense, hence infinite, and we conclude that $\operatorname{depth}(M_\mathfrak{p}) < s$ by Local Cohomology, Lemma 0DWY. In fact, let $V(\mathfrak{p}) \cap B = \{\mathfrak{p}_\lambda\}_{\lambda \in \Lambda}$. Pick $\mathfrak{q}_\lambda \in V(\mathfrak{p}_\lambda) \cap V(J) \cap V(I)$ as in (3). Let $\delta : \operatorname{Spec}(A) \to \mathbf{Z}$ be the dimension function associated to a dualizing complex $\omega_A^\bullet$ for A. Since Λ is infinite and δ is bounded, there exists an infinite subset $\Lambda' \subset \Lambda$ on which $\delta(\mathfrak{q}_\lambda)$ is constant. For $\lambda \in \Lambda'$ we have

$$\operatorname{depth}(M_{\mathfrak{p}_\lambda}) + \delta(\mathfrak{p}_\lambda) - \delta(\mathfrak{q}_\lambda) = \operatorname{depth}(M_{\mathfrak{p}_\lambda}) + \dim((A/\mathfrak{p}_\lambda)_{\mathfrak{q}_\lambda}) \leq d + s$$

by (3) and the definition of B. By the semi-continuity of the function $\operatorname{depth} + \delta$ proved in Duality for Schemes, Lemma 0ECM we conclude that

$$\operatorname{depth}(M_\mathfrak{p}) + \dim((A/\mathfrak{p})_{\mathfrak{q}_\lambda}) = \operatorname{depth}(M_\mathfrak{p}) + \delta(\mathfrak{p}) - \delta(\mathfrak{q}_\lambda) \leq d + s$$

Since also $\mathfrak{p} \notin V(I)$ we read off from (4) that $\mathfrak{p} \in T$, i.e., $V(\mathfrak{p}) \cap V(I) \subset V(J) \cap V(I)$. This finishes the proof of Claim I.

Claim II: $H^i_{J_0}(M) \to H^i_J(M)$ is an isomorphism for $i \leq s$ and $J' \subset J_0$ with $V(J') \cap V(I) = V(J) \cap V(I)$.

Proof of claim II. Choose $\mathfrak{p} \in V(J')$ not in $V(J_0)$. It suffices to show that $H^i_{\mathfrak{p}A_\mathfrak{p}}(M_\mathfrak{p}) = 0$ for $i \leq s$, see Local Cohomology, Lemma 0DWV. Observe that $\mathfrak{p} \in T$. Hence since $\mathfrak{p}$ is not in B we see that $\text{depth}(M_\mathfrak{p}) > s$ and the groups vanish by Dualizing Complexes, Lemma 0AVZ.

Claim III. The final statement of the lemma is true.

By Claim II for $i \leq s$ we have

$$H^i_T(M) = H^i_{J_0}(M) = H^i_{J'}(M)$$

for all ideals $J' \subset J_0$ with $V(J') \cap V(I) = V(J) \cap V(I)$. See Local Cohomology, Lemma 0EF1. Let us check the hypotheses of Local Cohomology, Proposition 0EFC for the subsets $T \subset T \cup V(I)$, the module M, and the integer s. We have to show that given $\mathfrak{p} \subset \mathfrak{q}$ with $\mathfrak{p} \notin T \cup V(I)$ and $\mathfrak{q} \in T$ we have

$$\text{depth}_{A_\mathfrak{p}}(M_\mathfrak{p}) + \dim((A/\mathfrak{p})_\mathfrak{q}) > s$$

If $\text{depth}(M_\mathfrak{p}) \geq s$, then this is true because the dimension of $(A/\mathfrak{p})_\mathfrak{q}$ is at least 1. Thus we may assume $\text{depth}(M_\mathfrak{p}) < s$. If $\mathfrak{q} \in V(I)$, then $\mathfrak{q} \in V(J) \cap V(I)$ and the inequality holds by (4). If $\mathfrak{q} \notin V(I)$, then we can use (3) to pick $\mathfrak{q}' \in V(\mathfrak{q}) \cap V(J) \cap V(I)$ with $\dim((A/\mathfrak{q})_{\mathfrak{q}'}) \leq d$. Then assumption (4) gives

$$\text{depth}_{A_\mathfrak{p}}(M_\mathfrak{p}) + \dim((A/\mathfrak{p})_{\mathfrak{q}'}) > s + d$$

Since A is catenary this implies the inequality we want. Applying Local Cohomology, Proposition 0EFC we find $J'' \subset A$ with $V(J'') \subset T \cup V(I)$ such that J'' annihilates $H^i_T(M)$ for $i \leq s$. Then we can write $V(J'') \cup V(J_0) \cup V(I) = V(J'I)$ for some $J' \subset J_0$ with $V(J') \cap V(I) = V(J) \cap V(I)$. Replacing J_0 by J' the proof is complete. $\square$